

Explainable AI for Academic Integrity Verification

Intermediate presentation: progress, writing, implementation

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The problem

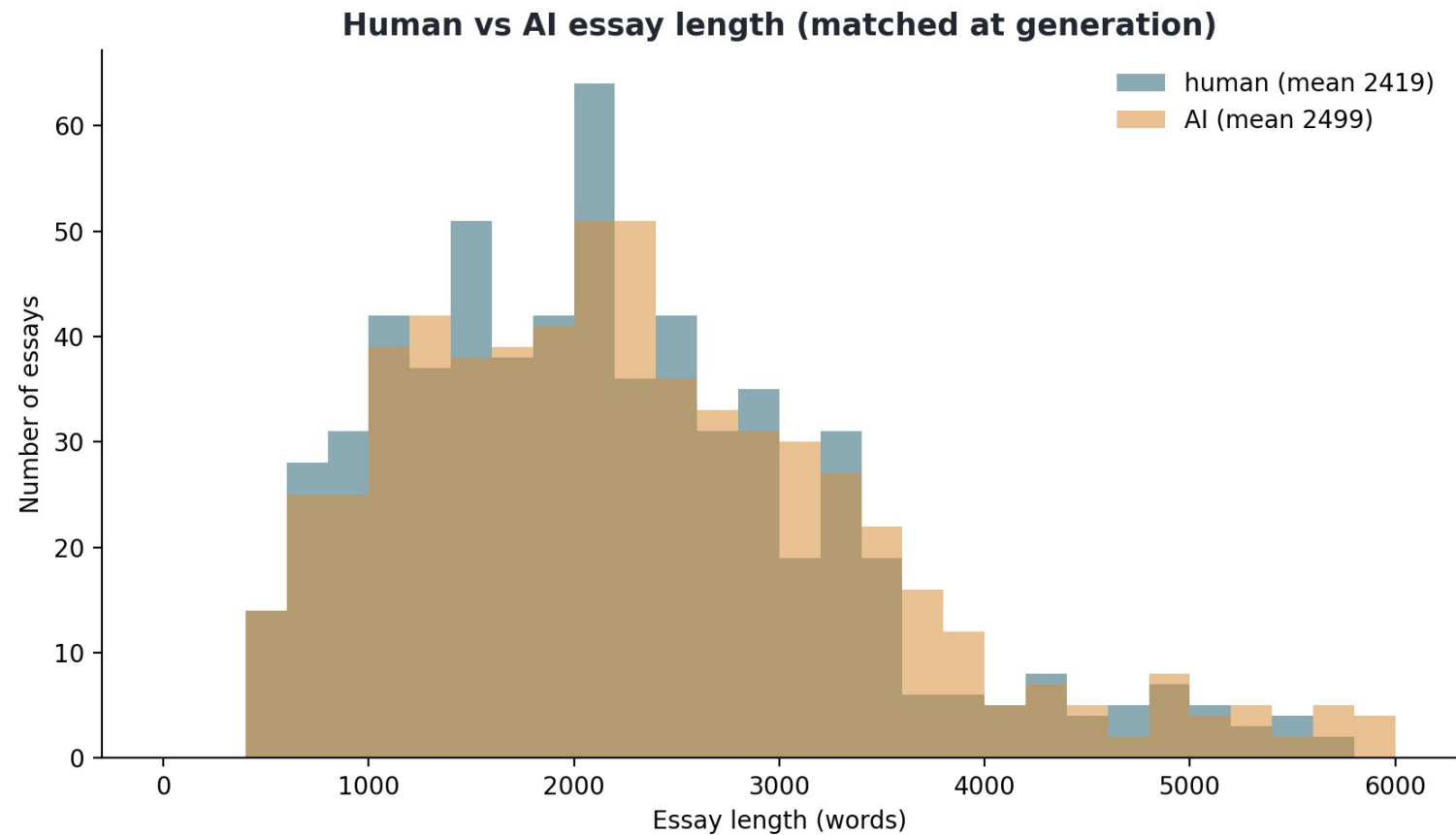
- Students use generative AI; universities lean on detectors that return one opaque percentage.
- A lecturer cannot defend '82% AI' if a student challenges it, and cannot tell whether it is right.
- The errors are unfair: detectors flag non-native English writers far more often (Liang et al. 2023: 61% vs 5% false-positive rate).
- Even a correct flag answers the wrong question. It says how the text was made, not whether the student understands it.

Research question and the system

- How can an explainable pipeline combine transparent AI-text detection with argument-aware question generation, and how far can a locally fine-tuned open model match a commercial LLM?
- Detect: a fine-tuned transformer plus interpretable stylometric features.
- Explain: show which features drove each decision, and test those explanations are faithful.
- Verify: extract the student's own claims (with provenance) and generate questions only the author could answer well, tagged by Bloom's level.
- Compare: the same generation task on a commercial API and a local open model.

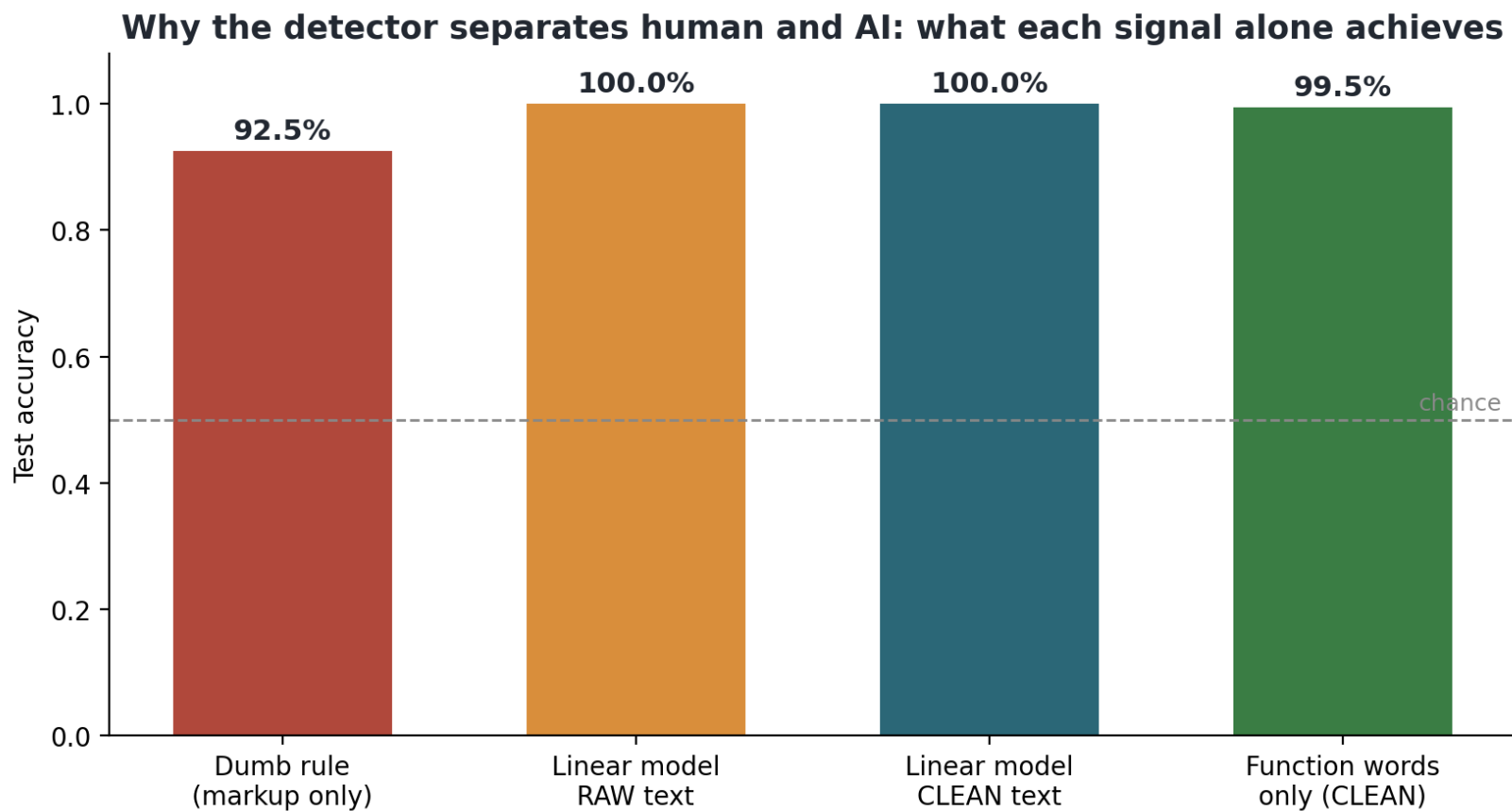
Implementation: dataset built and validated

640 human essays (BAWE, balanced across disciplines and native/non-native writers) matched with 640 AI essays generated locally, on the same topics at the same lengths. 1,280 essays; the length curves coincide, so the detector cannot cheat on length.



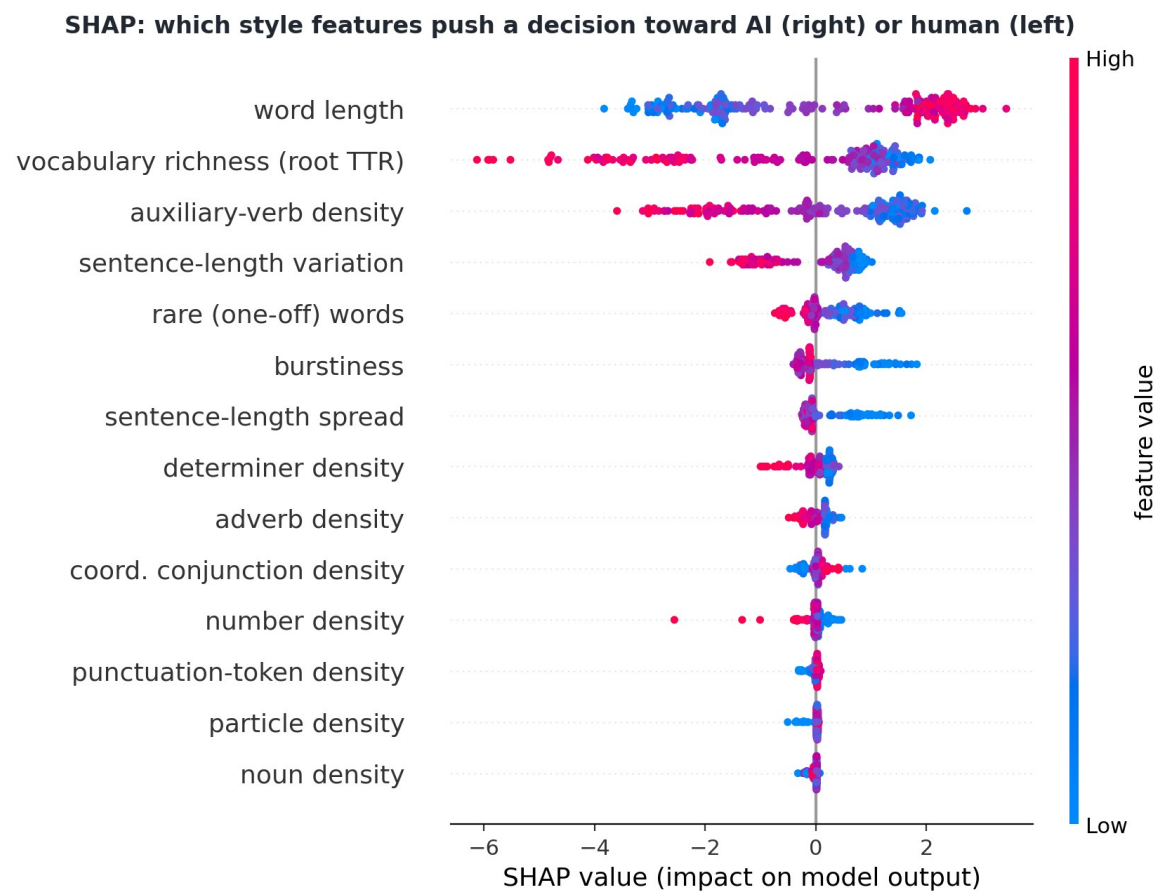
The detector, audited rather than trusted

The first model scored 100%: treated as a warning. The audit found corpus formatting tags a 'has-a-tag' rule alone scores 92.5% on; I stripped them from both classes and retrained. Cleaned results: DeBERTa F1 0.990, RoBERTa 0.995, and function words alone still reach 0.995, so the signal is genuine style.



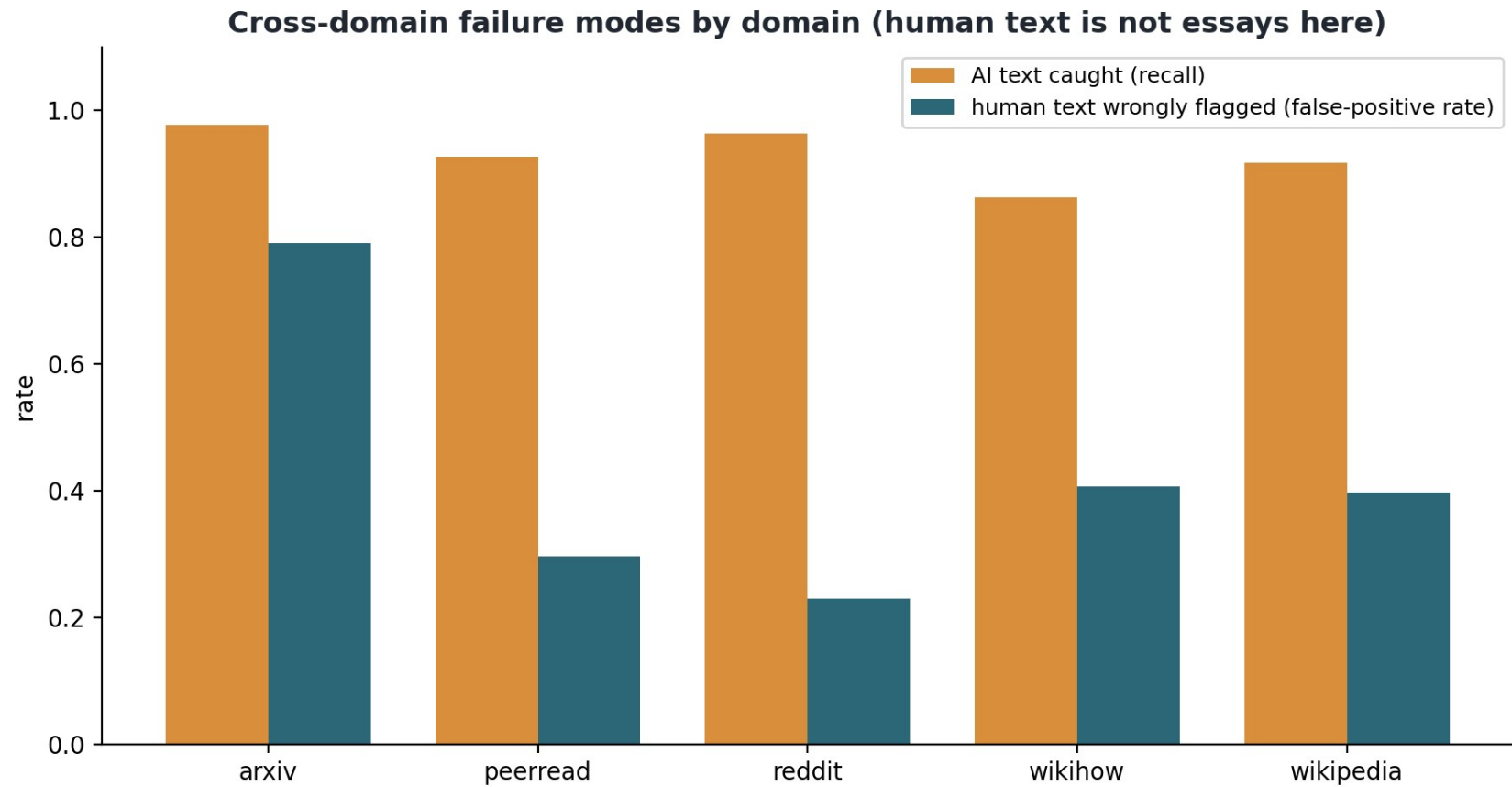
Explaining decisions a lecturer can defend

A style-only detector (F1 0.985) explained with SHAP: longer words push toward AI; richer vocabulary and varied sentence length push toward human. Tested for faithfulness by ablation (ERASER-style): the transformer's token attributions are diffuse and weakly faithful, so the style explanation is the one shown to lecturers.



Robustness: where it works and where it breaks

On unseen generators (OUTFOX essays in the M4 corpus) F1 holds at 0.97, catching GPT-4 and ChatGPT it never trained on. On unseen domains F1 drops to 0.79, and the failure is false accusation: real human arXiv abstracts flagged as AI 79% of the time. The fairness risk, measured.



From flag to verification questions

The Verification Interview Guide: claims extracted from the flagged essay, each cited to its exact sentence numbers (the quote is looked up by index, so the model cannot invent a source), with questions aimed at the author's own reasoning, tagged by Bloom's level.

Claim (cited to sentences 20, 36, 42 of the submission): the author argues that language is a powerful tool that conveys meaning, evokes emotion, and shapes understanding.

Q1. What specific examples from the texts you analyzed did you use to illustrate how language can convey meaning?

Bloom: understand

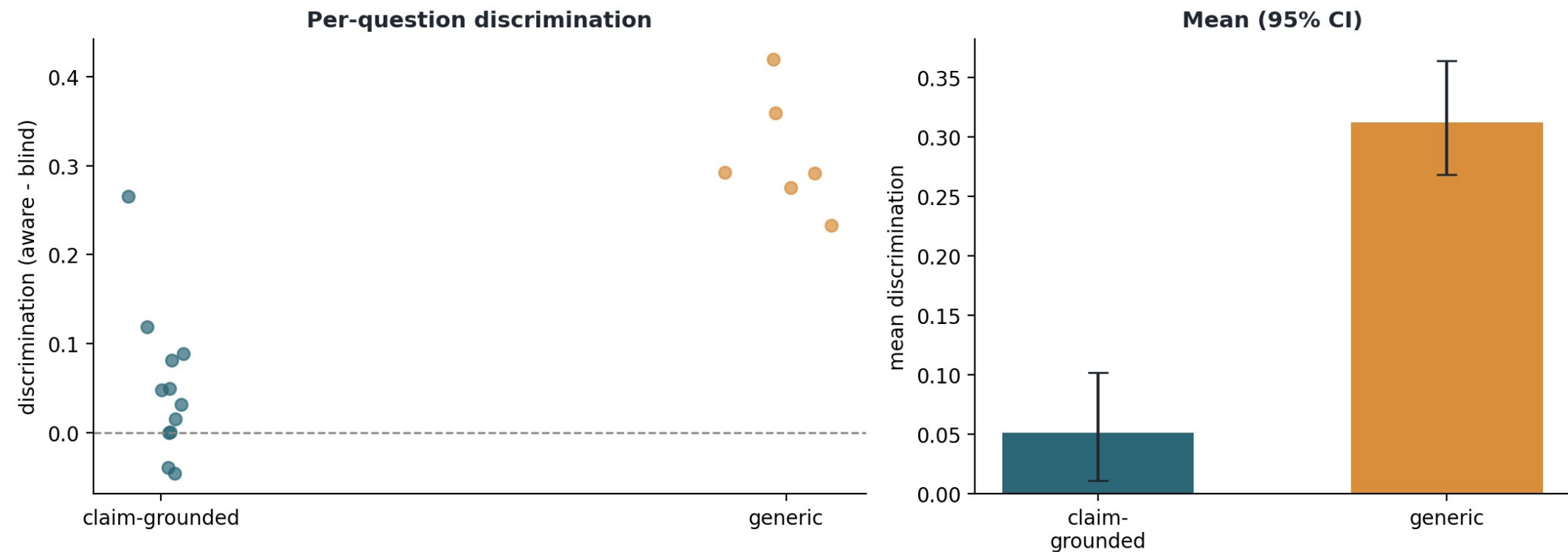
Q2. In your essay, you mentioned that social relationships are often marked by ambiguity and uncertainty. What particular aspect of the texts' use of language led you to make this claim, and how does it relate to your broader argument about the power of language?

Bloom: analyse

Evaluating the questions without a judge

Discrimination simulation: a model answers each question with the source and without it; the gap is the score. Surprise finding on the pilot essay: generic 'explain your own argument' questions discriminate more (0.31) than the claim-grounded ones (0.05), because a knowledgeable model answers specific questions from world knowledge.

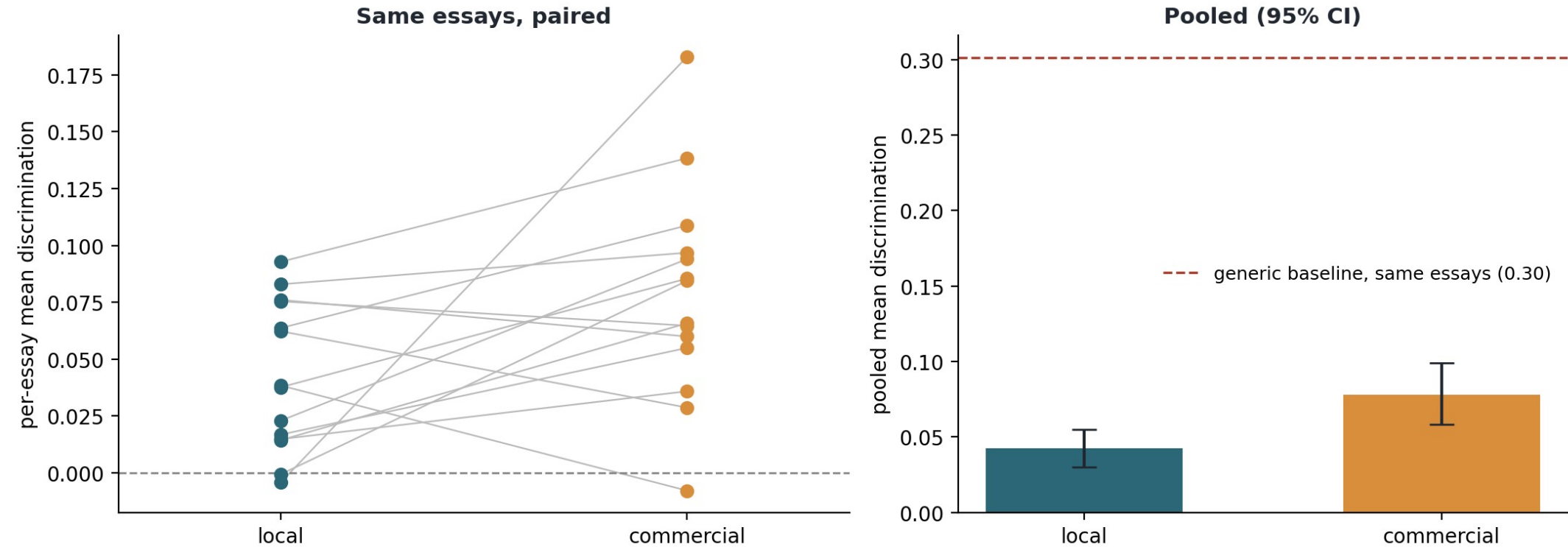
Discrimination simulation: claim-grounded vs generic verification questions



Commercial vs local: the first scaled answer

Same 14 essays, same scorer; only the question-writing model differs (126 questions each). Local Llama 3.1 8B 0.042 [0.030, 0.055] vs free-tier Gemini 0.078 [0.058, 0.098]. Paired difference 0.036 (t p = 0.040, Wilcoxon p = 0.030): a small but significant commercial edge, and the local model is competitive before any fine-tuning.

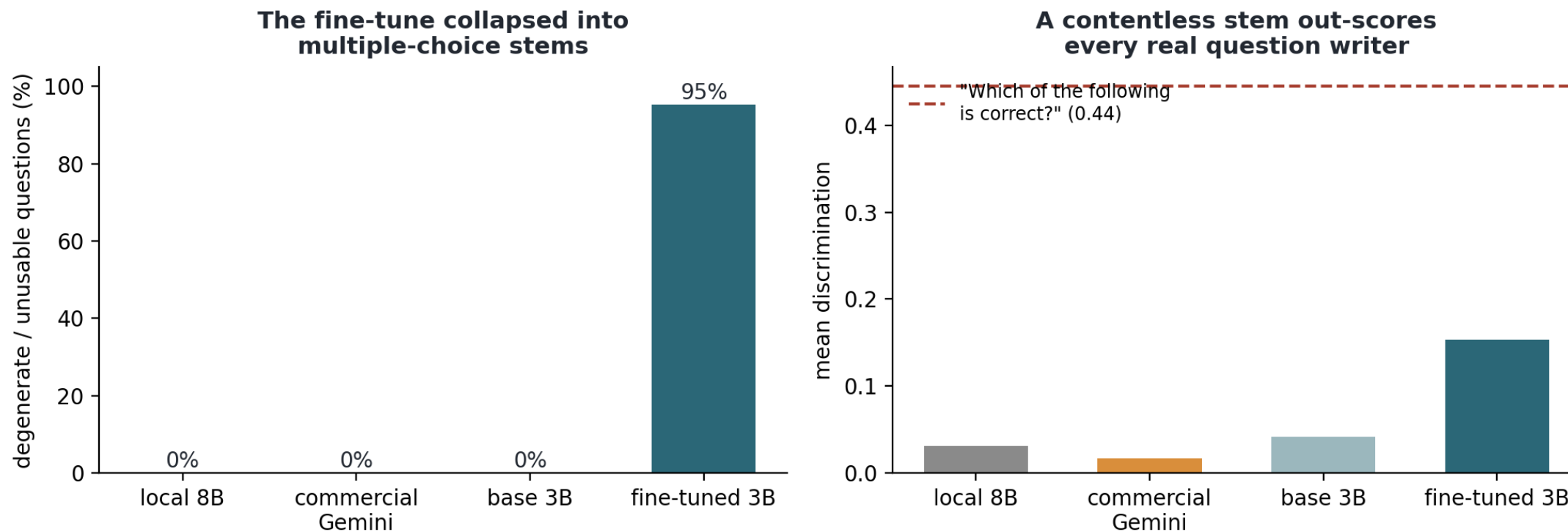
Question generation: commercial vs local across 14 essays



A result that looked too good: reading the questions

On a fixed claim set the QLoRA-fine-tuned 3B posted the highest discrimination by far (0.153 vs 0.024-0.041), seeming to beat commercial Gemini. Then I read its questions: 95% are degenerate multiple-choice stems like 'Which of the following is correct?' with no options. Those empty stems GAME the metric (that literal string scores 0.44, above every real question writer), because a contentless question makes the source-aware and source-blind answers diverge at random. The apparent win is an artifact; the fine-tune failed.

Why the fine-tuned model's discrimination score is an artifact



Writing: drafted as the work happened

- 56-page dissertation draft in the official 2026 ATU template: eight chapters plus a verified reference list.
- Chapter 1 rewritten to the supervisor's feedback: overview before background, elaborated background, expanded problem statement.
- Chapter 2 literature review written as prose, on 24 verified sources across the introduction and review (each checked against the real paper: arXiv, ACL Anthology, dblp, publisher).
- Results chapters (3 to 8) written while each component was fresh: dataset, audit, explainability, robustness, question generation, evaluation.
- Progress log kept per session; it feeds the methodology chapter and these slides.

Plan to submission (end of August)

- Bloom's classifier: already trained (BERT-base on EduQG's 903 labelled questions; doubles the keyword baseline, macro-F1 0.31 vs 0.16; the skewed label supply, 73% one class, is the bottleneck).
- Claim extractor: v1 already trained (DeBERTa on Persuasive Essays, official split; strict span-F1 0.63, premises 0.72).
Weeks 7 to 8: integrate spans-for-provenance with prompted phrasing, design settled with my supervisor.
- QLoRA fine-tune, complete as a three-way data-format experiment: v1 on EduQG collapsed (95% degenerate stems gaming the metric); v2 on SQuAD fixed the degeneracy with a real gain (0.102 vs base 0.027); v3 on verification questions self-distilled from the pipeline's own prompt writes on-style, fully well-formed questions at 0.064 (2.5x base, double its 8B teacher). The metric alone would rank v2 first despite unfit-for-purpose output, so it does not get to decide alone. On fixed claims the small open models match free-tier Gemini.
- Output assembler: first version already working end to end (live detector score with stated limits, claims with provenance, trained Bloom tags, suggested rubric, rendered to PDF).
- LLM-as-judge: complete for all three judges (Gemini, Claude, GPT). They disagree with each other (Krippendorff's alpha -0.25) and none tracks the objective simulation, so the judge-free measure carries the evidence, exactly as the evaluation plan anticipated. Weeks 9 to 10: polish the guide with lecturer-facing feedback.
- From early August: protected writing time; if anything slips, analysis depth is cut, not write-up.
- Ethics: no human participants (no ethics approval needed, confirmed with my supervisor); all datasets public and licensed (BAWE is CC BY-NC-SA); the output is verification evidence for a conversation, never an accusation.